ELECTRÓNICA DIGITAL 3

CLASE 3

Módulo Interno NVIC

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Módulo SCS

Dentro del mapa de memoria se encuentra una región denominada *SCS (System Control Space)* encargada del control de diversas funcionalidades del sistema:

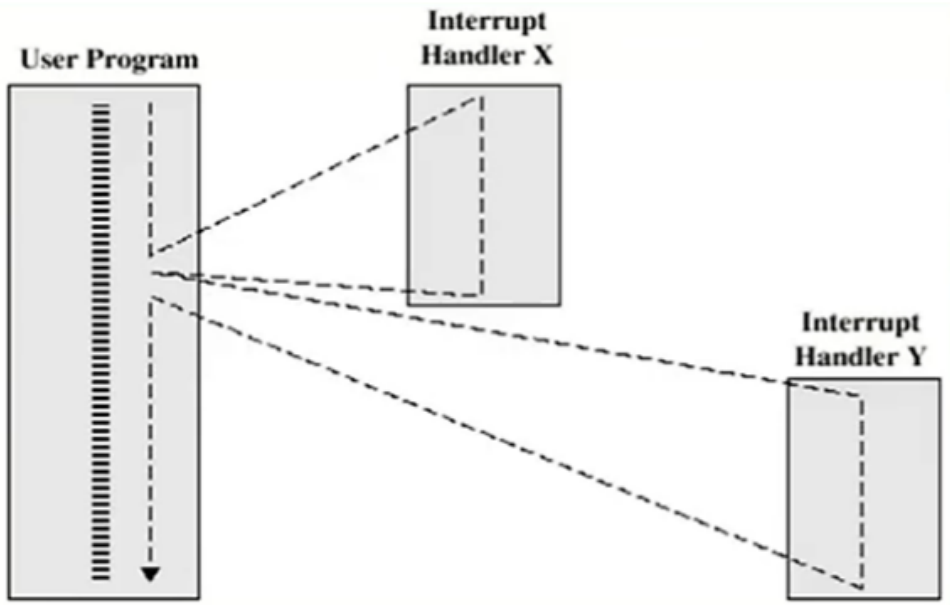
- **SCB (System Control Block):** Se encarga fundamentalmente del manejo excepciones del sistema.
  - Manejo de la tabla o vector de excepciones.
  - Habilitación e inhabilitación de excepciones.
  - Estado de las excepciones (inactiva, pendiente, activa)
  - Prioridades de las excepciones.
- **NVIC (Nested Vectored Interrupt Controller):** Se encarga de gestionar las interrupciones.
- **SysTick Timer:** Se encarga de gestionar el temporizador integrado en la arquitectura Cortex-M.

Módulo NVIC

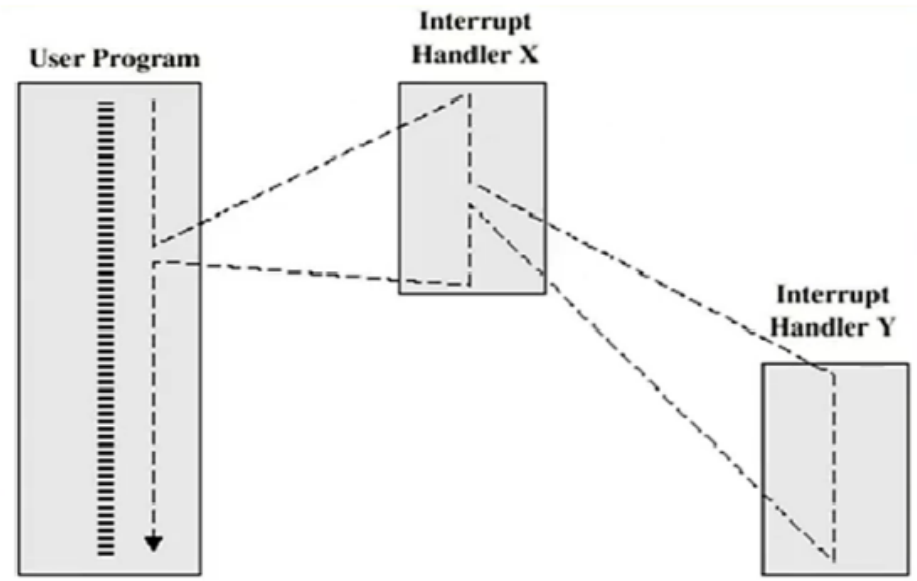
El NVIC (Nested Vectored Interrupt Control) es un Controlador de Interrupción Vectorial Anidado diseñado por ARM que se ubica dentro del core del microcontrolador.

Concepto de Anidamiento

El NVIC permite atender interrupciones anidadas, es decir interrupciones adentro de otras interrupciones.



Interrupcion Secuencial PIC
(Se puede atender solo una interrupción)



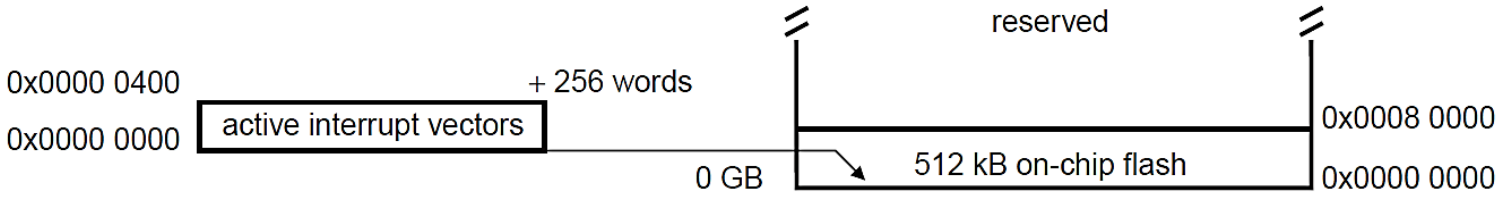
Interrupcion Anidada ARM
(Se puede atender multiples interrupciones)

Módulo NVIC

Concepto de Vectorización

En ARM las interrupciones se manejan con una tabla de vectores de interrupción que contiene una matriz de punteros de función. Cada puntero corresponde con un vector de interrupción, en donde se coloca la dirección en donde estará la rutina de servicio a la interrupción (IRQs) o Handler. Cada una de las direcciones está asociada a un periférico determinado.

En ARM Cortex M la tabla de vectores de interrupción tiene 256 posiciones de memoria reubicables y seteadas de fábrica. Dado que cada palabra tiene 4 bytes, se obtiene un total de 64 vectores de interrupción.

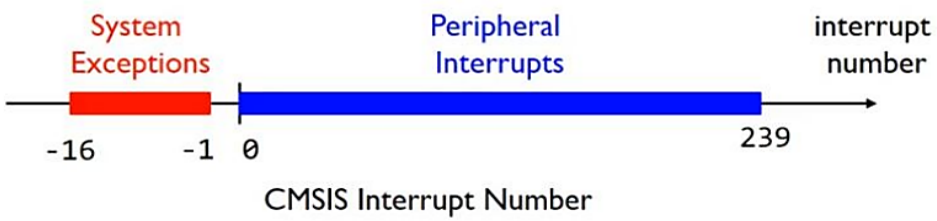


En ARM se tienen ***excepciones*** e ***interrupciones:***

*Las **excepciones**, se tratan de eventos que ocurren internamente durante la ejecución del programa y pueden abarcar una gama más amplia de situaciones. Esto incluye condiciones anómalas como errores de ejecución, intentos de acceder a memoria no válida, divisiones por cero, y otros eventos que pueden requerir intervención especial.*

*Las **interrupciones**, se tratan específicamente de eventos externos generados por periféricos u otros dispositivos que requieren la atención inmediata del procesador. Estos eventos externos provocan la suspensión temporal de la ejecución normal del programa y la transferencia del control a una rutina de manejo de interrupción específica.*

Módulo NVIC



En términos generales, la arquitectura Cortex-M3 admite hasta 256 vectores de interrupción y excepción. El NVIC del ARM Cortex M3 (ARM7RDMI) admite 35 interrupciones vectoriales (CMSIS considera las excepciones con numero negativo y las interrupciones con numero positivo).

Nota: Cada excepción tiene un nivel de prioridad, y sólo aquéllas de mayor prioridad pueden interrumpir a otra que esté siendo atendida. Los números más bajos tienen precedencia, es decir, una prioridad más alta.

Las excepciones RESET, NMI y HARDFAULT tienen las prioridades más altas, fijas e inamovibles, -3, -2 y -1 respectivamente. El resto de las excepciones tiene su prioridad configurable por software mediante registros en el SCB o el NVIC, comenzando con el valor 0 (la prioridad más alta).

Interrupt ID	Excepti on No.	Vector offset	Function	Interrupt ID	Excepti on No.	Vector offset	Function
0	16	0x40	Watchdog (WDT)	18	34	0x88	External interrupt
1	17	0x44	Timer 0	19	35	0x8C	External interrupt
2	18	0x48	Timer 1	20	36	0x90	External interrupt
3	19	0x4C	Timer 2	21	37	0x94	External interrupt
4	20	0x50	Timer 3	22	38	0x98	ADC
5	21	0x54	UART 0	23	39	0x9C	BOD
6	22	0x58	UART 1	24	40	0xA0	USB
7	23	0x5C	UART 2	25	41	0xA4	CAN
8	24	0x60	UART 3	26	42	0xA8	GPDMA
9	25	0x64	PWM 1	27	43	0xAC	I2S
10	26	0x68	I2C0	28	44	0xB0	Ethernet
11	27	0x6C	I2C1	29	45	0xB4	Repetitive int. timer
12	28	0x70	I2C2	30	46	0xB8	PWM
13	29	0x74	SPI	31	47	0xBC	Quadrature Encoder
14	30	0x78	SSP0	32	48	0xC0	PLL1 (USB PLL)
15	31	0x7C	SSP1	33	49	0xC4	USB activity int.
16	32	0x80	PLL0	34	50	0xC8	CAN activity int.
17	33	0x84	RTC				

Interrupt ID	Except ion No.	Priority	Vector offset	Function
	1	-3, highest	0x04	Reset
-14	2	-2	0x08	NMI
-13	3	-1	0x0C	Hard fault :
-12	4		0x10	Memory management fault : MPU violation , access to illegal locations
-11	5		0x14	Bus fault : AHB interface receives an error response from slave
-10	6		0x18	Usage fault : Program error or trying to access coprocessor
	7-10			Reserved
-5	11		0x2C	SVCall : System Service call
	12-13			Reserved
-2	14		0x38	PendSV : Pendable request for system device
-1	15		0x3C	SysTick : System tick timer

Módulo NVIC: Configuración Interrupciones por GPIO

Los siguientes registros configuran los pines del Puerto 0 y Puerto 2 para generar interrupciones.

Registro IO0IntEnR (GPIO Interrupt Enable for port 0 Rising Edge)

Cada bit en estos registros habilita la interrupción del flanco ascendente para el correspondiente pin del puerto 0.

GPIO Interrupt Enable for port 0 Rising Edge (IO0IntEnR - 0x4002 8090) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0ER		Enable rising edge interrupt for P0.0.	0
		0	Rising edge interrupt is disabled on P0.0.	
		1	Rising edge interrupt is enabled on P0.0.	
1	P0.1ER		Enable rising edge interrupt for P0.1.	0
2	P0.2ER		Enable rising edge interrupt for P0.2.	0
3	P0.3ER		Enable rising edge interrupt for P0.3.	0
4	P0.4ER[1]		Enable rising edge interrupt for P0.4.	0
5	P0.5ER[1]		Enable rising edge interrupt for P0.5.	0
6	P0.6ER		Enable rising edge interrupt for P0.6.	0
7	P0.7ER		Enable rising edge interrupt for P0.7.	0
8	P0.8ER		Enable rising edge interrupt for P0.8.	0
9	P0.9ER		Enable rising edge interrupt for P0.9.	0
10	P0.10ER		Enable rising edge interrupt for P0.10.	0
11	P0.11ER		Enable rising edge interrupt for P0.11.	0
14:12	-		Reserved	NA
15	P0.15ER		Enable rising edge interrupt for P0.15.	0
16	P0.16ER		Enable rising edge interrupt for P0.16.	0
17	P0.17ER		Enable rising edge interrupt for P0.17.	0
18	P0.18ER		Enable rising edge interrupt for P0.18.	0
19	P0.19ER[1]		Enable rising edge interrupt for P0.19.	0
20	P0.20ER[1]		Enable rising edge interrupt for P0.20.	0
21	P0.21ER[1]		Enable rising edge interrupt for P0.21.	0
22	P0.22ER		Enable rising edge interrupt for P0.22.	0
23	P0.23ER[1]		Enable rising edge interrupt for P0.23.	0
24	P0.24ER[1]		Enable rising edge interrupt for P0.24.	0
25	P0.25ER		Enable rising edge interrupt for P0.25.	0
26	P0.26ER		Enable rising edge interrupt for P0.26.	0
27	P0.27ER[1]		Enable rising edge interrupt for P0.27.	0
28	P0.28ER[1]		Enable rising edge interrupt for P0.28.	0
29	P0.29ER		Enable rising edge interrupt for P0.29.	0
30	P0.30ER		Enable rising edge interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO2IntEnR (GPIO Interrupt Enable for port 2 Rising Edge)

Cada bit en estos registros habilita la interrupción del flanco ascendente para el correspondiente pin del puerto 2.

GPIO Interrupt Enable for port 2 Rising Edge (IO2IntEnR - 0x4002 80B0) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0ER		Enable rising edge interrupt for P2.0.	0
		0	Rising edge interrupt is disabled on P2.0.	
		1	Rising edge interrupt is enabled on P2.0.	
1	P2.1ER		Enable rising edge interrupt for P2.1.	0
2	P2.2ER		Enable rising edge interrupt for P2.2.	0
3	P2.3ER		Enable rising edge interrupt for P2.3.	0
4	P2.4ER		Enable rising edge interrupt for P2.4.	0
5	P2.5ER		Enable rising edge interrupt for P2.5.	0
6	P2.6ER		Enable rising edge interrupt for P2.6.	0
7	P2.7ER		Enable rising edge interrupt for P2.7.	0
8	P2.8ER		Enable rising edge interrupt for P2.8.	0
9	P2.9ER		Enable rising edge interrupt for P2.9.	0
10	P2.10ER		Enable rising edge interrupt for P2.10.	0
11	P2.11ER[1]		Enable rising edge interrupt for P2.11.	0
12	P2.12ER[1]		Enable rising edge interrupt for P2.12.	0
13	P2.13ER[1]		Enable rising edge interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO0IntEnF (GPIO Interrupt Enable for port 0 Falling Edge)

Cada bit en estos registros habilita la interrupción del flanco descendente para el correspondiente pin del puerto 0.

GPIO Interrupt Enable for port 0 Falling Edge (IO0IntEnF - address 0x4002 8094) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0EF		Enable falling edge interrupt for P0.0	0
		0	Falling edge interrupt is disabled on P0.0.	
		1	Falling edge interrupt is enabled on P0.0.	
1	P0.1EF		Enable falling edge interrupt for P0.1.	0
2	P0.2EF		Enable falling edge interrupt for P0.2.	0
3	P0.3EF		Enable falling edge interrupt for P0.3.	0
4	P0.4EF[1]		Enable falling edge interrupt for P0.4.	0
5	P0.5EF[1]		Enable falling edge interrupt for P0.5.	0
6	P0.6EF		Enable falling edge interrupt for P0.6.	0
7	P0.7EF		Enable falling edge interrupt for P0.7.	0
8	P0.8EF		Enable falling edge interrupt for P0.8.	0
9	P0.9EF		Enable falling edge interrupt for P0.9.	0
10	P0.10EF		Enable falling edge interrupt for P0.10.	0
11	P0.11EF		Enable falling edge interrupt for P0.11.	0
14:12	-		Reserved.	NA

Bit	Symbol	Value	Description	Reset value
15	P0.15EF		Enable falling edge interrupt for P0.15.	0
16	P0.16EF		Enable falling edge interrupt for P0.16.	0
17	P0.17EF		Enable falling edge interrupt for P0.17.	0
18	P0.18EF		Enable falling edge interrupt for P0.18.	0
19	P0.19EF[1]		Enable falling edge interrupt for P0.19.	0
20	P0.20EF[1]		Enable falling edge interrupt for P0.20.	0
21	P0.21EF[1]		Enable falling edge interrupt for P0.21.	0
22	P0.22EF		Enable falling edge interrupt for P0.22.	0
23	P0.23EF[1]		Enable falling edge interrupt for P0.23.	0
24	P0.24EF[1]		Enable falling edge interrupt for P0.24.	0
25	P0.25EF		Enable falling edge interrupt for P0.25.	0
26	P0.26EF		Enable falling edge interrupt for P0.26.	0
27	P0.27EF[1]		Enable falling edge interrupt for P0.27.	0
28	P0.28EF[1]		Enable falling edge interrupt for P0.28.	0
29	P0.29EF		Enable falling edge interrupt for P0.29.	0
30	P0.30EF		Enable falling edge interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO2IntEnF (GPIO Interrupt Enable for port 2 Falling Edge)

Cada bit en estos registros habilita la interrupción del flanco descendente para el correspondiente pin del puerto 2.

GPIO Interrupt Enable for port 2 Falling Edge (IO2IntEnF - 0x4002 80B4) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0EF		Enable falling edge interrupt for P2.0	0
		0	Falling edge interrupt is disabled on P2.0.	
		1	Falling edge interrupt is enabled on P2.0.	
1	P2.1EF		Enable falling edge interrupt for P2.1.	0
2	P2.2EF		Enable falling edge interrupt for P2.2.	0
3	P2.3EF		Enable falling edge interrupt for P2.3.	0
4	P2.4EF		Enable falling edge interrupt for P2.4.	0
5	P2.5EF		Enable falling edge interrupt for P2.5.	0
6	P2.6EF		Enable falling edge interrupt for P2.6.	0
7	P2.7EF		Enable falling edge interrupt for P2.7.	0
8	P2.8EF		Enable falling edge interrupt for P2.8.	0
9	P2.9EF		Enable falling edge interrupt for P2.9.	0
10	P2.10EF		Enable falling edge interrupt for P2.10.	0
11	P2.11EF[1]		Enable falling edge interrupt for P2.11.	0
12	P2.12EF[1]		Enable falling edge interrupt for P2.12.	0
13	P2.13EF[1]		Enable falling edge interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO0IntStatR (GPIO Interrupt Status for port 0 Rising Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco ascendente para el puerto 0.

GPIO Interrupt Status for port 0 Rising Edge Interrupt (IO0IntStatR - 0x4002 8084) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0REI		Status of Rising Edge Interrupt for P0.0	0
		0	A rising edge has not been detected on P0.0.	
		1	Interrupt has been generated due to a rising edge on P0.0.	
1	P0.1REI		Status of Rising Edge Interrupt for P0.1.	0
2	P0.2REI		Status of Rising Edge Interrupt for P0.2.	0
3	P0.3REI		Status of Rising Edge Interrupt for P0.3.	0
4	P0.4REI[1]		Status of Rising Edge Interrupt for P0.4.	0
5	P0.5REI[1]		Status of Rising Edge Interrupt for P0.5.	0
6	P0.6REI		Status of Rising Edge Interrupt for P0.6.	0
7	P0.7REI		Status of Rising Edge Interrupt for P0.7.	0
8	P0.8REI		Status of Rising Edge Interrupt for P0.8.	0
9	P0.9REI		Status of Rising Edge Interrupt for P0.9.	0
10	P0.10REI		Status of Rising Edge Interrupt for P0.10.	0
11	P0.11REI		Status of Rising Edge Interrupt for P0.11.	0
14:12	-		Reserved.	NA
15	P0.15REI		Status of Rising Edge Interrupt for P0.15.	0
16	P0.16REI		Status of Rising Edge Interrupt for P0.16.	0
17	P0.17REI		Status of Rising Edge Interrupt for P0.17.	0
18	P0.18REI		Status of Rising Edge Interrupt for P0.18.	0
19	P0.19REI[1]		Status of Rising Edge Interrupt for P0.19.	0
20	P0.20REI[1]		Status of Rising Edge Interrupt for P0.20.	0
21	P0.21REI[1]		Status of Rising Edge Interrupt for P0.21.	0
22	P0.22REI		Status of Rising Edge Interrupt for P0.22.	0
23	P0.23REI[1]		Status of Rising Edge Interrupt for P0.23.	0
24	P0.24REI[1]		Status of Rising Edge Interrupt for P0.24.	0
25	P0.25REI		Status of Rising Edge Interrupt for P0.25.	0
26	P0.26REI		Status of Rising Edge Interrupt for P0.26.	0
27	P0.27REI[1]		Status of Rising Edge Interrupt for P0.27.	0
28	P0.28REI[1]		Status of Rising Edge Interrupt for P0.28.	0
29	P0.29REI		Status of Rising Edge Interrupt for P0.29.	0
30	P0.30REI		Status of Rising Edge Interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO2IntStatR (GPIO Interrupt Status for port 2 Rising Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco ascendente para el puerto 2.

GPIO Interrupt Status for port 2 Rising Edge Interrupt (IO2IntStatR - 0x4002 80A4) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0REI		Status of Rising Edge Interrupt for P2.0	0
		0	A rising edge has not been detected on P2.0.	
		1	Interrupt has been generated due to a rising edge on P2.0.	
1	P2.1REI		Status of Rising Edge Interrupt for P2.1.	0
2	P2.2REI		Status of Rising Edge Interrupt for P2.2.	0
3	P2.3REI		Status of Rising Edge Interrupt for P2.3.	0
4	P2.4REI		Status of Rising Edge Interrupt for P2.4.	0
5	P2.5REI		Status of Rising Edge Interrupt for P2.5.	0
6	P2.6REI		Status of Rising Edge Interrupt for P2.6.	0
7	P2.7REI		Status of Rising Edge Interrupt for P2.7.	0
8	P2.8REI		Status of Rising Edge Interrupt for P2.8.	0
9	P2.9REI		Status of Rising Edge Interrupt for P2.9.	0
10	P2.10REI		Status of Rising Edge Interrupt for P2.10.	0
11	P2.11REI[1]		Status of Rising Edge Interrupt for P2.11.	0
12	P2.12REI[1]		Status of Rising Edge Interrupt for P2.12.	0
13	P2.13REI[1]		Status of Rising Edge Interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO0IntStatF (GPIO Interrupt Status for port 0 Falling Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco descendente para el puerto 0.

GPIO Interrupt Status for port 0 Falling Edge Interrupt (IO0IntStatF - 0x4002 8088) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0FEI		Status of Falling Edge Interrupt for P0.0	0
		0	A falling edge has not been detected on P0.0.	
		1	Interrupt has been generated due to a falling edge on P0.0.	
1	P0.1FEI		Status of Falling Edge Interrupt for P0.1.	0
2	P0.2FEI		Status of Falling Edge Interrupt for P0.2.	0
3	P0.3FEI		Status of Falling Edge Interrupt for P0.3.	0
4	P0.4FEI[1]		Status of Falling Edge Interrupt for P0.4.	0
5	P0.5FEI[1]		Status of Falling Edge Interrupt for P0.5.	0
6	P0.6FEI		Status of Falling Edge Interrupt for P0.6.	0
7	P0.7FEI		Status of Falling Edge Interrupt for P0.7.	0
8	P0.8FEI		Status of Falling Edge Interrupt for P0.8.	0
9	P0.9FEI		Status of Falling Edge Interrupt for P0.9.	0
10	P0.10FEI		Status of Falling Edge Interrupt for P0.10.	0
11	P0.11FEI		Status of Falling Edge Interrupt for P0.11.	0
14:12	-		Reserved.	NA
15	P0.15FEI		Status of Falling Edge Interrupt for P0.15.	0
16	P0.16FEI		Status of Falling Edge Interrupt for P0.16.	0
17	P0.17FEI		Status of Falling Edge Interrupt for P0.17.	0
18	P0.18FEI		Status of Falling Edge Interrupt for P0.18.	0
19	P0.19FEI[1]		Status of Falling Edge Interrupt for P0.19.	0
20	P0.20FEI[1]		Status of Falling Edge Interrupt for P0.20.	0
21	P0.21FEI[1]		Status of Falling Edge Interrupt for P0.21.	0
22	P0.22FEI		Status of Falling Edge Interrupt for P0.22.	0
23	P0.23FEI[1]		Status of Falling Edge Interrupt for P0.23.	0
24	P0.24FEI[1]		Status of Falling Edge Interrupt for P0.24.	0
25	P0.25FEI		Status of Falling Edge Interrupt for P0.25.	0
26	P0.26FEI		Status of Falling Edge Interrupt for P0.26.	0
27	P0.27FEI[1]		Status of Falling Edge Interrupt for P0.27.	0
28	P0.28FEI[1]		Status of Falling Edge Interrupt for P0.28.	0
29	P0.29FEI		Status of Falling Edge Interrupt for P0.29.	0
30	P0.30FEI		Status of Falling Edge Interrupt for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO2IntStatF (GPIO Interrupt Status for port 2 Falling Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco descendente para el puerto 2.

GPIO Interrupt Status for port 2 Falling Edge Interrupt (IO2IntStatF - 0x4002 80A8) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0FEI		Status of Falling Edge Interrupt for P2.0	0
		0	A falling edge has not been detected on P2.0.	
		1	Interrupt has been generated due to a falling edge on P2.0.	
1	P2.1FEI		Status of Falling Edge Interrupt for P2.1.	0
2	P2.2FEI		Status of Falling Edge Interrupt for P2.2.	0
3	P2.3FEI		Status of Falling Edge Interrupt for P2.3.	0
4	P2.4FEI		Status of Falling Edge Interrupt for P2.4.	0
5	P2.5FEI		Status of Falling Edge Interrupt for P2.5.	0
6	P2.6FEI		Status of Falling Edge Interrupt for P2.6.	0
7	P2.7FEI		Status of Falling Edge Interrupt for P2.7.	0
8	P2.8FEI		Status of Falling Edge Interrupt for P2.8.	0
9	P2.9FEI		Status of Falling Edge Interrupt for P2.9.	0
10	P2.10FEI		Status of Falling Edge Interrupt for P2.10.	0
11	P2.11FEI[1]		Status of Falling Edge Interrupt for P2.11.	0
12	P2.12FEI[1]		Status of Falling Edge Interrupt for P2.12.	0
13	P2.13FEI[1]		Status of Falling Edge Interrupt for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO0IntClr (GPIO Interrupt Clear register for port 0)

Escribir un 1 en un bit en este registro borra cualquier interrupción para el correspondiente pin del puerto 0.

GPIO Interrupt Clear register for port 0 (IO0IntClr - 0x4002 808C)) bit description

Bit	Symbol	Value	Description	Reset value
0	P0.0CI		Clear GPIO port Interrupts for P0.0	0
		0	Corresponding bits in IOxIntStatR and IOxIntStatF are unchanged.	
		1	Corresponding bits in IOxIntStatR and IOxStatF are cleared.	
1	P0.1CI		Clear GPIO port Interrupts for P0.1.	0
2	P0.2CI		Clear GPIO port Interrupts for P0.2.	0
3	P0.3CI		Clear GPIO port Interrupts for P0.3.	0
4	P0.4CI[1]		Clear GPIO port Interrupts for P0.4.	0
5	P0.5CI[1]		Clear GPIO port Interrupts for P0.5.	0
6	P0.6CI		Clear GPIO port Interrupts for P0.6.	0
7	P0.7CI		Clear GPIO port Interrupts for P0.7.	0
8	P0.8CI		Clear GPIO port Interrupts for P0.8.	0
9	P0.9CI		Clear GPIO port Interrupts for P0.9.	0
10	P0.10CI		Clear GPIO port Interrupts for P0.10.	0
11	P0.11CI		Clear GPIO port Interrupts for P0.11.	0
14:12	-		Reserved.	NA
15	P0.15CI		Clear GPIO port Interrupts for P0.15.	0
16	P0.16CI		Clear GPIO port Interrupts for P0.16.	0
17	P0.17CI		Clear GPIO port Interrupts for P0.17.	0
18	P0.18CI		Clear GPIO port Interrupts for P0.18.	0
19	P0.19CI[1]		Clear GPIO port Interrupts for P0.19.	0
20	P0.20CI[1]		Clear GPIO port Interrupts for P0.20.	0
21	P0.21CI[1]		Clear GPIO port Interrupts for P0.21.	0
22	P0.22CI		Clear GPIO port Interrupts for P0.22.	0
23	P0.23CI[1]		Clear GPIO port Interrupts for P0.23.	0
24	P0.24CI[1]		Clear GPIO port Interrupts for P0.24.	0
25	P0.25CI		Clear GPIO port Interrupts for P0.25.	0
26	P0.26CI		Clear GPIO port Interrupts for P0.26.	0
27	P0.27CI[1]		Clear GPIO port Interrupts for P0.27.	0
28	P0.28CI[1]		Clear GPIO port Interrupts for P0.28.	0
29	P0.29CI		Clear GPIO port Interrupts for P0.29.	0
30	P0.30CI		Clear GPIO port Interrupts for P0.30.	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IO2IntClr (GPIO Interrupt Clear register for port 2)

Escribir un 1 en un bit en este registro borra cualquier interrupción para el correspondiente pin del puerto 2.

GPIO Interrupt Clear register for port 2 (IO2IntClr - 0x4002 80AC) bit description

Bit	Symbol	Value	Description	Reset value
0	P2.0CI		Clear GPIO port Interrupts for P2.0	0
		0	Corresponding bits in IOxIntStatR and IOxIntStatF are unchanged.	
		1	Corresponding bits in IOxIntStatR and IOxStatF are cleared.	
1	P2.1CI		Clear GPIO port Interrupts for P2.1.	0
2	P2.2CI		Clear GPIO port Interrupts for P2.2.	0
3	P2.3CI		Clear GPIO port Interrupts for P2.3.	0
4	P2.4CI		Clear GPIO port Interrupts for P2.4.	0
5	P2.5CI		Clear GPIO port Interrupts for P2.5.	0
6	P2.6CI		Clear GPIO port Interrupts for P2.6.	0
7	P2.7CI		Clear GPIO port Interrupts for P2.7.	0
8	P2.8CI		Clear GPIO port Interrupts for P2.8.	0
9	P2.9CI		Clear GPIO port Interrupts for P2.9.	0
10	P2.10CI		Clear GPIO port Interrupts for P2.10.	0
11	P2.11CI[1]		Clear GPIO port Interrupts for P2.11.	0
12	P2.12CI[1]		Clear GPIO port Interrupts for P2.12.	0
13	P2.13CI[1]		Clear GPIO port Interrupts for P2.13.	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

Módulo NVIC: Configuración Interrupciones por GPIO

Registro IOIntStatus (GPIO overall Interrupt Status register)

Este registro indica la presencia de interrupción pendiente en todos los puertos que admiten interrupciones GPIO. Solo se requiere un bit de estado por puerto.

GPIO overall Interrupt Status register (IOIntStatus - address 0x4002 8080) bit description

Bit	Symbol	Value	Description	Reset value
0	P0Int		Port 0 GPIO interrupt pending.	0
		0	There are no pending interrupts on Port 0.	
		1	There is at least one pending interrupt on Port 0.	
1	-	-	Reserved. The value read from a reserved bit is not defined.	NA
2	P2Int		Port 2 GPIO interrupt pending.	0
		0	There are no pending interrupts on Port 2.	
		1	There is at least one pending interrupt on Port 2.	
31:2	-	-	Reserved. The value read from a reserved bit is not defined.	NA

Módulo NVIC: Configuración de Interrupciones Externas

Registro EXTINT (External Interrupt Flag register)

External Interrupt Flag register (EXTINT - address 0x400F C140) bit description

Bit	Symbol	Description	Reset value
0	EINT0	In level-sensitive mode, this bit is set if the EINT0 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT0 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.[1]	0
1	EINT1	In level-sensitive mode, this bit is set if the EINT1 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT1 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.[1]	0
2	EINT2	In level-sensitive mode, this bit is set if the EINT2 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT2 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.[1]	0
3	EINT3	In level-sensitive mode, this bit is set if the EINT3 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT3 function is selected for its pin, and the selected edge occurs on the pin. This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.[1]	0
31:4	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

[1] Example: e.g. if the EINTx is selected to be low level sensitive and low level is present on corresponding pin, this bit can not be cleared; this bit can be cleared only when signal on the pin becomes high.

Módulo NVIC: Configuración de Interrupciones Externas

Registro EXTMODE (External Interrupt Mode register)

Los bits en este registro seleccionan si cada pin EINT es sensible a una interrupción por nivel o por flanco.

External Interrupt Mode register (EXTMODE - address 0x400F C148) bit description

Bit	Symbol	Value	Description	Reset value
0	EXTMODE0	0	Level-sensitivity is selected for $\overline{\text{EINT0}}$.	0
		1	$\overline{\text{EINT0}}$ is edge sensitive.	
1	EXTMODE1	0	Level-sensitivity is selected for $\overline{\text{EINT1}}$.	0
		1	$\overline{\text{EINT1}}$ is edge sensitive.	
2	EXTMODE2	0	Level-sensitivity is selected for $\overline{\text{EINT2}}$.	0
		1	$\overline{\text{EINT2}}$ is edge sensitive.	
3	EXTMODE3	0	Level-sensitivity is selected for $\overline{\text{EINT3}}$.	0
		1	$\overline{\text{EINT3}}$ is edge sensitive.	
31:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Módulo NVIC: Configuración de Interrupciones Externas

Registro EXTPOLAR (External Interrupt Polarity register)

En el modo sensible por nivel, los bits en este registro seleccionan si el pin correspondiente es activo por alto o bajo. En el modo sensible por flanco, seleccionan si el pin es activo por flanco ascendente o por flanco descendente.

External Interrupt Polarity register (EXTPOLAR - address 0x400F C14C) bit description

Bit	Symbol	Value	Description	Reset value
0	EXTPOLAR0	0	$\overline{\text{EINT0}}$ is low-active or falling-edge sensitive (depending on EXTMODE0).	0
		1	$\overline{\text{EINT0}}$ is high-active or rising-edge sensitive (depending on EXTMODE0).	
1	EXTPOLAR1	0	$\overline{\text{EINT1}}$ is low-active or falling-edge sensitive (depending on EXTMODE1).	0
		1	$\overline{\text{EINT1}}$ is high-active or rising-edge sensitive (depending on EXTMODE1).	
2	EXTPOLAR2	0	$\overline{\text{EINT2}}$ is low-active or falling-edge sensitive (depending on EXTMODE2).	0
		1	$\overline{\text{EINT2}}$ is high-active or rising-edge sensitive (depending on EXTMODE2).	
3	EXTPOLAR3	0	$\overline{\text{EINT3}}$ is low-active or falling-edge sensitive (depending on EXTMODE3).	0
		1	$\overline{\text{EINT3}}$ is high-active or rising-edge sensitive (depending on EXTMODE3).	
31:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Módulo NVIC: Funciones de Control de Interrupciones por CMSIS

Las funciones para configurar interrupciones son:

CMSIS functions for NVIC control

CMSIS interrupt control function	Description
`void NVIC_SetPriorityGrouping(uint32_t priority_grouping)`	Set the priority grouping
`void NVIC_EnableIRQ(IRQn_t IRQn)`	Enable IRQn
`void NVIC_DisableIRQ(IRQn_t IRQn)`	Disable IRQn
`uint32_t NVIC_GetPendingIRQ (IRQn_t IRQn)`	Return true (IRQ-Number) if IRQn is pending
`void NVIC_SetPendingIRQ (IRQn_t IRQn)`	Set IRQn pending
`void NVIC_ClearPendingIRQ (IRQn_t IRQn)`	Clear IRQn pending status
`uint32_t NVIC_GetActive (IRQn_t IRQn)`	Return the IRQ number of the active interrupt
`void NVIC_SetPriority (IRQn_t IRQn, uint32_t priority)`	Set priority for IRQn
`uint32_t NVIC_GetPriority (IRQn_t IRQn)`	Read priority of IRQn
`void NVIC_SystemReset (void)`	Reset the system

Ejercicio 3.3: Interrupción por GPIO

Realizar un programa que configure el puerto P2.0 y P2.1 para que provoquen una interrupción por flanco de subida para el primer pin y por flanco de bajada para el segundo. Cuando la interrupción sea por P2.0 se enviará por el pin P0.0 la secuencia de bits 010011010. Si la interrupción es por P2.1 se enviará por el pin P0.1 la secuencia 011100110. Las secuencias se envían únicamente cuando se produce una interrupción, en caso contrario la salida de los pines tienen valores 1 lógicos. ¿que prioridad tienen configuradas por defecto estas interrupciones?

Ejercicio 3.4: Interrupción por GPIO

Realizar un programa que configure el puerto P0.0 y P2.0 para que provoquen una interrupción por flanco de subida. Si la interrupción es por P0.0 guardar el valor binario 100111 en la variable "auxiliar", si es por P2.0 guardar el valor binario 111001011010110.